
Reliable Donation-Commitment Labeling and Multi-Label Persuasion Strategy Modeling

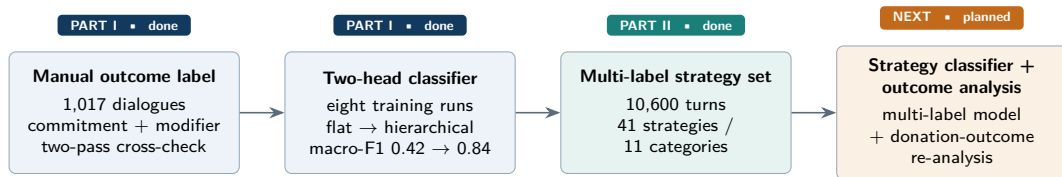
PersuasionForGood: from a corrected outcome label to strategy combinations

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Group 10 ▪ Capstone Final Presentation

Project roadmap



Part I — the corpus's own donation field is unreliable, so we rebuilt the label from the transcript and learned it.

Part II — a binary outcome alone cannot explain *why* persuasion works, so we added turn-level strategy sets.

Part I — the ground truth disagrees with the transcript

PersuasionForGood ships 1,017 dialogues with a numeric donation field that essentially *every* downstream study treats as ground truth.

It is demonstrably wrong in identifiable cases

- A dialogue where the Persuadee explicitly confirms a \$10 donation is recorded as **\$0**.
- “I will absolutely look to donating something later”
→ both numeric fields show **\$0**.

Two compounding problems

- **Field vs. text** — the label is not derived from what was said.
- **Binary is too coarse** — a firm “yes” and a hedged or conditional “yes” collapse into the same bucket.

Research question: can a manually annotated, text-grounded commitment label be built *and* learned well enough to replace the corpus field?

Corpus field

recorded donation = \$0.00



What the transcript says

“we have you down for a \$10 donation. Is that correct?”
“yes that is” ■ *“yes thats ok”*



Our replacement

commitment = yes ■ modifier = (blank)

Building the label — annotation scheme

Two fields per dialogue, from the text only

- `binary_label` $\in \{\text{yes}, \text{no}\}$ — did the Persuadee make an *unconditional* commitment to donate within *this* task?
- `modifier` $\in \{\text{blank}, \text{conditional}, \text{deferred}\}$ — captures commitments the numeric field cannot represent at all.

Rules of thumb

- Use only what the Persuadee *explicitly* says.
- Take their *last clear position*.
- The Persuader's own claims never count.
- No clear answer $\Rightarrow$ no, blank modifier.

Worked examples from the guideline

What the Persuadee says	label	modifier
"Yes, \$10 sounds good, go ahead."	yes	—
"No thanks, I'm not interested."	no	—
"I'd donate \$2 if you guarantee it goes to local kids." (<i>never confirmed</i>)	no	cond.
"I'll check their website later and maybe donate there."	no	def.
"Sure, \$1 now, and I might give more later."	yes	def.

Five annotators, disjoint partitions, one shared guideline $\rightarrow$ **1,017 labelled dialogues**.

Building the label — reliability check

An independent **second annotation pass** over 154 dialogues, labelled strictly from the text with *no access* to the first pass.

92.21 %

two-pass agreement
(142 of 154 dialogues)

A further 28 borderline calls were flagged as cases where a different labeller could reasonably differ, even where the two passes agreed.

The 12 disagreements were not random — they cluster into four recurring boundary cases:

- **Hedged but unretracted** phrasing — “I may give a few cents”.
- The **conditional / deferred boundary** — is the condition the Persuadee’s own future action, or something the Persuader must confirm?
- **Persuadee match-offers** — does “I’ll match your donation” constitute a firm commitment?
- **Off-topic or malformed** task instances.

These four patterns were written back into the guideline *before* the remaining partitions were labelled — tightened against observed failure modes, not rewritten.

Dataset I — distribution, and a normalization step that mattered

Raw annotator output was not directly usable — casing and whitespace drifted across the five annotators:

commitment label	count
yes / Yes	635 / 103
no / No / "No "	229 / 49 / 1

modifier	count
none / blank	732 / 139
deferred / Deferred	106 / 19
conditional / Conditional	19 / 2

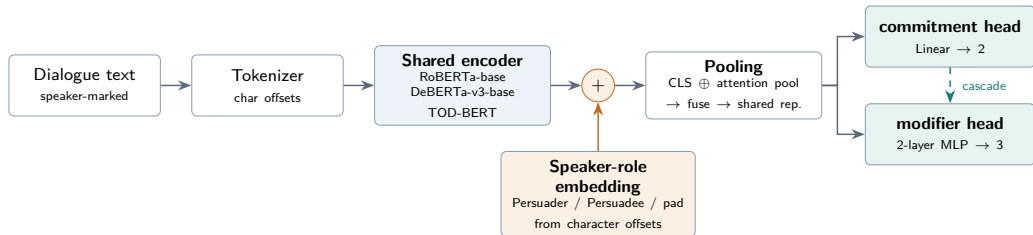
A naive grouping would have silently split yes from Yes into two different classes.

After normalization — the dataset actually used

Class	Count	Share
<i>commitment</i>		
yes	738	72.6 %
no	279	27.4 %
<i>modifier</i>		
blank (none)	871	85.6 %
deferred	125	12.3 %
conditional	21	2.1 %

The headline risk for modelling: conditional has only **21** examples in the entire corpus — about 3 in a 153-row test split. Every number reported for that class has to be read against this.

The classifier — shared encoder, two heads



Two design choices that matter

- **Speaker-role embedding** — added to the hidden states before pooling, so the encoder knows which spans are the Persuadee's. Only that speaker's words define the label.
- **Cascaded modifier head** — it receives the commitment head's own softmax output, matching the task's conditional structure (a modifier only qualifies a commitment).

Evaluation protocol

- Stratified 70 / 15 / 15 split on the joint commitment × modifier stratum (711 / 153 / 153).
- Majority-class baseline computed on the *same* test split.
- Headline metric is **macro-F1**. A constant predictor already scores 0.841 positive-class F1 here; macro-F1 puts that baseline at **0.420**.

Early runs — the commitment head improves, the modifier head does not

What we tried, in order

- **Numerical stabilisation** — DeBERTa-v3 produced NaN losses; traced to the optimiser epsilon, plus a lower learning rate.
- **Speaker-role encoding** added to the pooled representation.
- **Attention pooling, focal loss and a supervised-contrastive auxiliary** replacing mean-pooling and weighted cross-entropy.
- **An augmentation-integrity repair.** 57.5 % of speaker tags are glued to a newline, so whitespace tokenisation never saw them — 15.9 % of augmented rows had corrupted markers. A regex tokeniser took that to 0 %.
- **A second augmentation round** — connective paraphrase and French back-translation for the rare classes.

The commitment head responded: every encoder cleared the majority baseline; speaker roles added +2–9 pp.

Commitment head — macro-F1 (test set)

Encoder	baseline encoder	+ speaker roles	+ augment. rounds
majority		0.420	
RoBERTa	0.680	0.700	0.648
DeBERTa-v3	0.530	0.620	0.670
TOD-BERT	0.590	0.670	0.623

But the modifier head never became reliable. $F_1(\text{conditional}) = 0.00$ in every run. And deferred was erratic — in every one of these flat runs *at least one* of the three encoders collapsed to 0.00, and none exceeded 0.29.

More augmentation did not fix it — at one point it made things worse. Four flat runs, no configuration in which all three encoders learned the class. That pattern points at the *data path*, not the architecture.

The diagnosis — the model never saw the answer

We stopped tuning and measured the data path instead. Three facts, all computed on the real 1,017-dialogue corpus with the actual tokenizer:

1. Dialogues are **long** — median **523** tokens, mean 548, max 1,528. **99.5 %** exceed the 256-token input budget in use.
2. The label-bearing span sits at the **end** — the Persuadee's final turn begins at **95.6 %** through the dialogue on average, and past the 60 % mark in **100 %** of dialogues.
3. The tokenizer truncates from the **right** by default, keeping the *first* 256 tokens.

Consequence: **1,010 of 1,017** dialogues (**99.3 %**) had their ending cut off. The model was trained *and* scored on text that does not contain the commitment — so it could only learn the class prior. That is exactly what a dead modifier head and a near-baseline commitment head look like.

What the model read

the first 256 tokens:
greetings, small talk, the pitch

discarded

Where the label lives

the Persuadee's closing turn:
"yes that is" ▪ "I'll donate \$1"

Two independent fixes, both tested

- A Keep the *end* of the flat sequence — **truncate from the left**, with a longer input budget.
- B Stop flattening at all — a **hierarchical encoder** that reads the dialogue turn by turn.

Fix A — keep the dialogue's ending

Three changes

- **Truncate from the left**, input budget 512 tokens. Dialogues that lose their ending drop from **1,010 / 1,017** to **0 / 1,017**.
- **Label-safe augmentation**. Earlier rounds perturbed the whole dialogue, Persuadee turns included — random deletion could remove the commitment itself, manufacturing mislabelled rows. Now **only the Persuader's turns are edited**. Since the guideline defines the label solely from Persuadee text, this is *provably* label-preserving, and it is asserted at run time.
- **Modifier trained on all rows**. **45 %** of informative modifiers (65 of 146) sit on no rows, exactly as the guideline specifies. Restricting the head to yes rows discarded them.

Commitment macro-F1 / deferred F_1

Encoder	before	keeping the ending
RoBERTa	0.648 / 0.09	0.842 / 0.07
DeBERTa-v3	0.670 / 0.13	0.421 / 0.00
TOD-BERT	0.623 / 0.00	0.774 / 0.15

RoBERTa reaches 0.842 macro-F1 (accuracy 0.882) — double the majority baseline, and the best commitment score of the whole project. Simply keeping the end of the text did that.

One failure worth reporting: DeBERTa-v3 *collapsed* to predicting yes for all 153 test rows (macro-F1 0.421 — exactly the baseline). The long flat sequence is unstable for it, which is part of why Fix B matters.

Fix B — hierarchical dialogue encoder



Each turn is encoded *independently* — batched as one flattened forward pass rather than a Python loop — and a small Transformer then runs over the sequence of turn vectors.

Dialogues have at most **30 turns** (95th percentile 23), so a 32×64 budget covers **over 98 %** of dialogues in full. The flat model's core defect is removed by construction rather than patched.

Training objective (shared with the later flat runs)

- **Class-balanced focal loss** on both heads — re-weights by effective number of samples and down-weights easy examples.
- **Supervised contrastive auxiliary** on the pooled dialogue representation, pulling same-class dialogues together.
- Gradient checkpointing and reduced batch sizes to fit the turn-flattened forward pass in memory.

Commitment head — macro-F1

Encoder	hierarchical	+ augmentation
RoBERTa	0.793	0.794
DeBERTa-v3	0.797	0.821
TOD-BERT	0.807	0.773

Modifier head — deferred F_1

Encoder	hierarchical	+ augmentation
RoBERTa	0.26	0.25
DeBERTa-v3	0.23	0.18
TOD-BERT	0.16	0.29

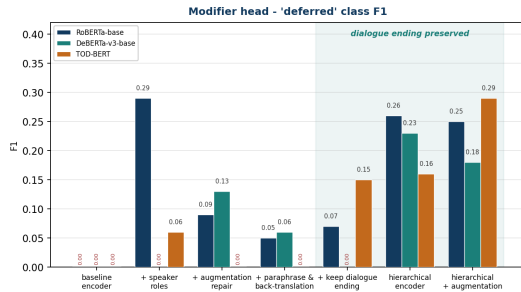
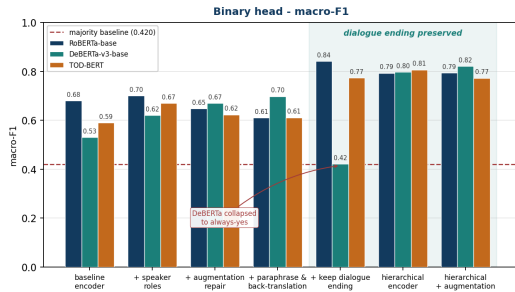
Robustness is the story here. All three encoders land at **0.77–0.82** macro-F1 with *no* collapse — the consistency Fix A lacked, where one encoder became a constant predictor.

Why combining helps: keeping turns separate lets the back-translated *turn* paraphrases feed the dialogue Transformer directly — augmentation the flat model could not exploit.

The modifier head finally holds. For the first time *all three* encoders clear zero on deferred at once (**0.16–0.29**); every flat run lost at least one to 0.00.

Still unsolved: conditional F_1 stays at ≈ 0.00 (best 0.07). With **21** examples corpus-wide and **3** in test, this is a data limit, not a modelling one.

Results across all eight training runs



Commitment head: majority baseline **0.420** → **0.842**. Roughly *double* the baseline.

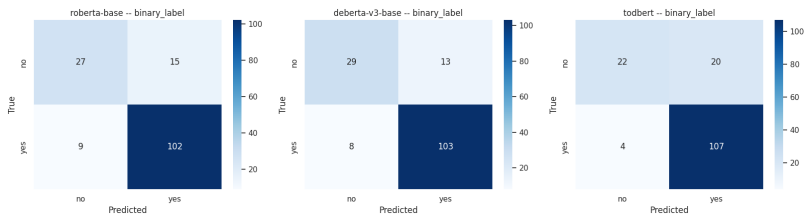
Modifier head: only the hierarchical runs get *all three* encoders above zero on deferred at once (**0.16–0.29**); every flat run lost at least one to 0.00.

The decisive change was neither the loss nor the augmentation — it was letting the model *read the end of the dialogue*.

Detailed results — commitment head

Best configuration (hierarchical encoder + augmentation) ■ test set, 153 dialogues

Encoder	no (<i>n</i> =42)			yes (<i>n</i> =111)			Acc	Macro- <i>F</i> ₁
	P	R	<i>F</i> ₁	P	R	<i>F</i> ₁		
RoBERTa-base	0.75	0.64	0.69	0.87	0.92	0.89	0.84	0.79
DeBERTa-v3-base	0.78	0.69	0.73	0.89	0.93	0.91	0.86	0.82
TOD-BERT	0.85	0.52	0.65	0.84	0.96	0.90	0.84	0.77
<i>majority baseline</i>	—	0.00	0.00	0.73	1.00	0.84	0.73	0.42

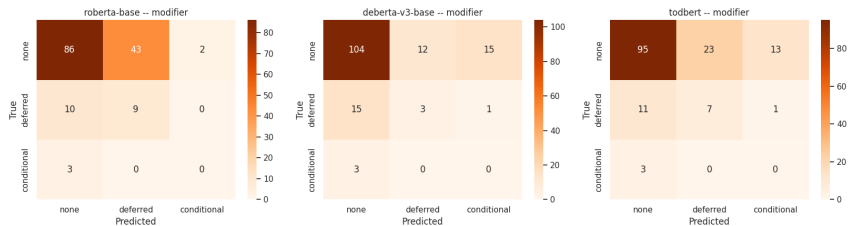


Reading the matrices: every encoder recovers yes strongly (recall 0.92–0.96); the residual error is no→yes false positives (13–20 of 42). That is why *macro-F1* sits below accuracy — and why we report it.

Detailed results — modifier head

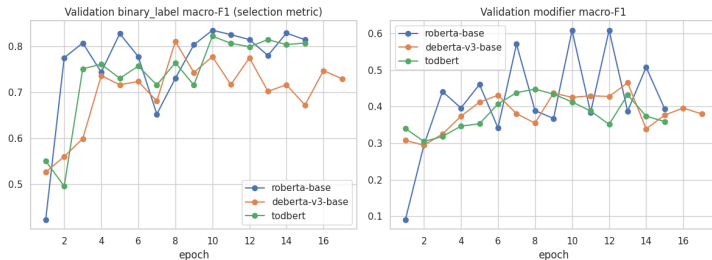
Same configuration ■ test set, 153 dialogues

Encoder	none ($n=131$)			deferred ($n=19$)			conditional ($n=3$)			Macro- F_1
	P	R	F_1	P	R	F_1	P	R	F_1	
RoBERTa-base	0.87	0.66	0.75	0.17	0.47	0.25	0.00	0.00	0.00	0.33
DeBERTa-v3-base	0.85	0.79	0.82	0.20	0.16	0.18	0.00	0.00	0.00	0.33
TOD-BERT	0.87	0.73	0.79	0.23	0.37	0.29	0.00	0.00	0.00	0.36



The bottom row of every matrix is identical: all 3 conditional dialogues are predicted none by all three encoders — the clearest single picture of the data limit.

Training dynamics and model selection



What the curves show

- The commitment head converges quickly and **stably** — all three encoders clear 0.75 within four epochs, then plateau.
- The modifier head is **visibly noisier**, swinging between 0.30 and 0.60 — that variance is the 19 deferred and 3 conditional validation rows.

How models were selected

- Selection on the **mean of both macro-F1 scores**, with early stopping and best-checkpoint reload — never the final epoch.
- The tuned threshold is reported *alongside* the untuned one: on 153 validation rows the tuner can overfit, and in one run it made test accuracy *worse*.

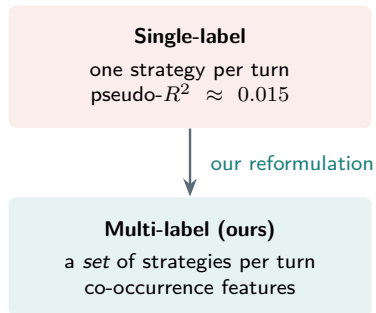
Part II — why we extended to persuasion strategies

Part I gives a *reliable outcome*. It does not explain **why** a dialogue ends in a donation.

The gap in the literature

- Wang et al. (2019) annotate 10 strategies — *one label per unit*.
- Petrova et al. (2026) re-annotate all 10,600 turns with a richer **41-strategy / 11-category** taxonomy — still *single-label*.
- Their own result: single-strategy presence barely predicts the outcome (**pseudo- $R^2 \approx 0.015$ – 0.08**), and they name *strategy combinations* as the open question.

Forcing one label per turn throws information away. “...around since 1919 and top rated by Charity Watch” is simultaneously an organizational-credibility claim *and* an evidence-and-statistics claim.



Our contribution. A turn-level *multi-label* strategy resource over the full corpus, so outcome modelling can use strategy *combinations* instead of a single forced pick — paired with our corrected outcome label.

The taxonomy — 11 categories, 41 strategies

Rational Appeal 4

- logical_appeal
- evidence_and_statistics
- cost_benefit_framing
- feasibility_and_ease

Emotional Appeal 4

- emotion_appeal
- guilt_induction
- empathy_and_perspective_taking
- hope_and_positive_impact

Credibility Appeal 4

- organizational_credibility
- transparency_and_accountability
- source_citation
- personal_credibility

Social Influence 4

- social_proof
- self_modeling
- in_group_appeal
- authority_endorsement

Reciprocity and Exchange 3

- reciprocity
- donor_benefit
- gratitude_and_appreciation

Commitment and Consistency 4

- foot_in_the_door
- door_in_the_face
- value_consistency_appeal
- incremental_ask

Framing and Presentation 4

- anchoring
- minimization_framing
- loss_versus_gain_framing
- comparison_framing

Urgency and Scarcity 3

- urgency_appeal
- scarcity_appeal
- deadline_pressure

Threat and Pressure 3

- negative_consequence_warning
- persistent_repetition
- obligation_pressure

Information Provision 4

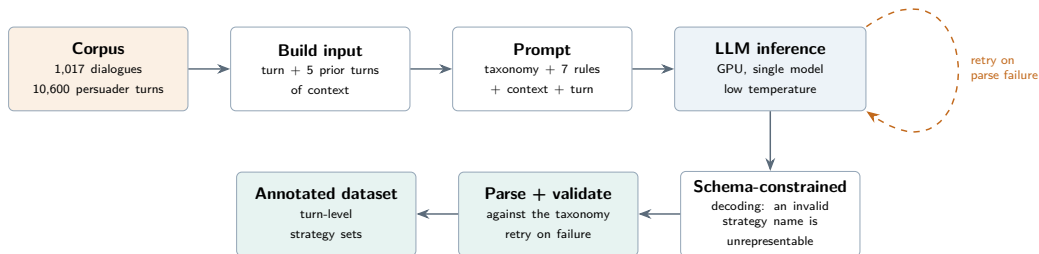
- donation_procedure_information
- organization_information
- impact_information
- task_clarification

Relational and Interactive 4

- personal_story
- personal_related_inquiry
- source_related_inquiry
- rapport_building

All 41 strategies and all 11 categories occur in the annotated corpus — none of the taxonomy is dead weight.

Annotation pipeline — 10,600 turns



Design choices

- A **single annotator model**, not a majority vote across three — a scope cut on cost and runtime, not on methodology.
- **Schema-constrained decoding** makes an invalid strategy name impossible to emit, not something caught afterwards.

Consequences we carry forward

- No cross-model agreement figure — a three-model design would have produced one.
- Work was split into **four batches**; batch identity is kept as a field, and its effect is shown later.

The annotation prompt

You are an expert annotator of persuasion strategies in dialogue.

In each conversation a **PERSUADER** tries to convince a **PERSUADEE** to donate part of their task payment to the charity Save the Children. You will be shown up to 5 prior turns of context and then **ONE** persuader turn.

Identify **EVERY** persuasion strategy the persuader turn uses, from the taxonomy below. A turn may use several strategies at once - this is a **MULTI-LABEL** task.

TAXONOMY (11 categories, 41 strategies)

<category> - <definition>

- <strategy>: <definition> e.g. "<cue phrase>"

Reply with JSON only:

```
{"labels": [{"category": "<category>",  
              "strategy": "<strategy>"}]}
```

The strategy must belong to the category you pair it with.

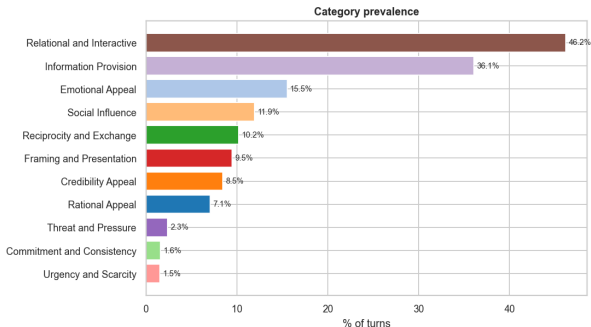
The seven rules

1. **Removal test** — two strategies are separate only if deleting the words carrying one still leaves the other.
2. **Specific beats general** — numbers used as proof are evidence, not *also* a logical appeal.
3. **Function, not vocabulary** — “children are starving” after a refusal is *also* guilt induction.
4. Most turns use **1–3**; more than six is disallowed.
5. An empty set *only* for genuinely non-persuasive turns.
6. Label the persuader turn only, never the context.
7. Label what is present, **not what worked** — the outcome is the dependent variable later, so leaking it would invalidate the analysis.

Dataset II — 10,600 annotated turns

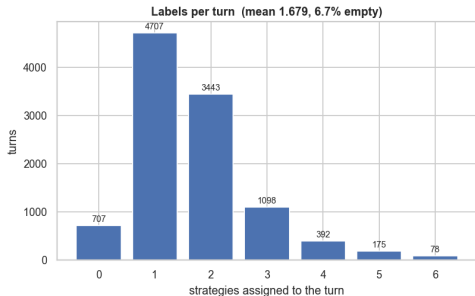
Metric	Value
Turns	10,600
Dialogues	1,017
Turns / dialogue	10.42
Mean labels / turn	1.68
Median	1
Empty-set turns	707 (6.7 %)
Multi-label (≥ 2)	48.9 %
Max labels	6
Strategies used	41 / 41
Categories used	11 / 11

The core justification: 48.9 % of turns carry two or more strategies. Nearly half the corpus was being flattened by the single-label formulation.

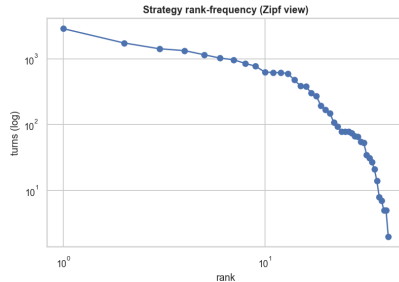


Relational and Interactive (46 %) and *Information Provision* (36 %) dominate; *Threat and Pressure*, *Commitment and Consistency* and *Urgency and Scarcity* each stay below 3 %.

Label distribution

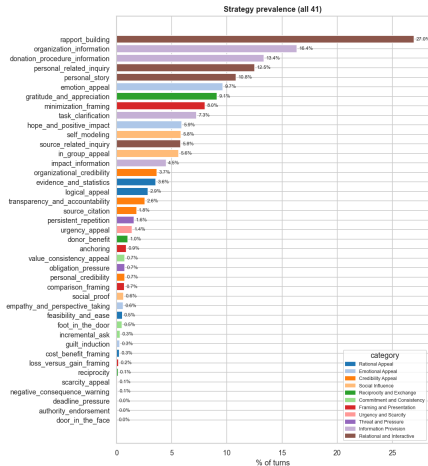


Single labels are the mode (44.4 %), but the tail is substantial — 32.5 % carry two, 10.4 % three, and 6.1 % four or more.



A Zipf-like rank/frequency curve: rapport building (27 %) dominates, while the long tail — door-in-the-face, deadline pressure, authority endorsement — fires on only a handful of turns. Those rare labels are the hard part of the next phase.

Strategy prevalence — all 41

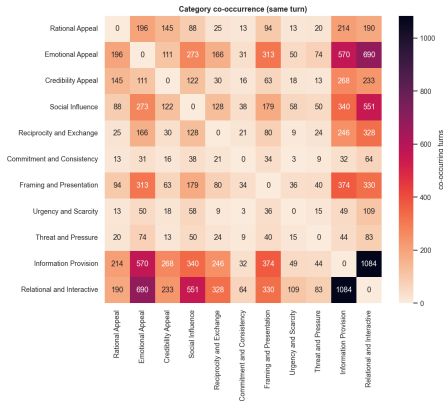


What the distribution shows

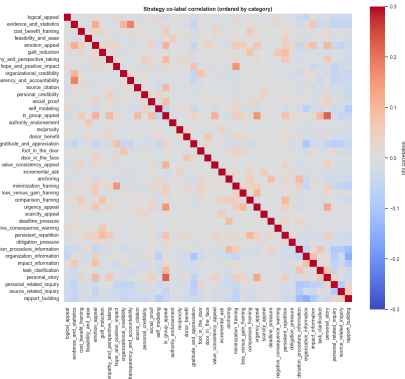
- The five commonest strategies are all **relational or informational** — rapport building, organization information, donation procedure, personal inquiry, personal story.
- Classic high-pressure tactics are **rare in practice**: guilt induction, scarcity, deadline pressure and door-in-the-face together account for under 1 % of turns.
- Persuaders here mostly build a relationship and supply information; coercive moves are the exception.

This long tail is why the next phase reports *per-label* scores — an average would hide the rare classes.

How strategies combine

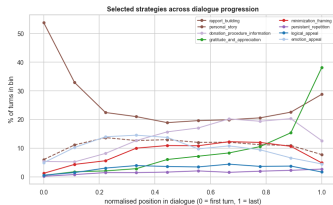


Category co-occurrence. Persuaders bundle *Relational* and *Information* moves with *Emotional* and *Social* appeals — rarely with cold *Rational* argument.

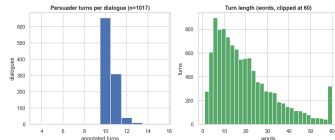


Strategy correlation. The same picture holds at the strategy level; the near-absence of strong negative correlations means strategies are *stacked*, not *substituted* — precisely what a single-label scheme cannot represent.

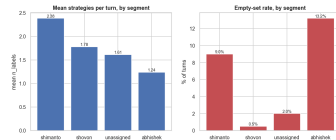
Structure across the dialogue, and an annotation caveat



Position. Strategies are *ordered*: rapport building and inquiries are front-loaded, while donation-procedure information and persistent repetition rise toward the close.



Length. Turns per dialogue and words per turn — the same length profile that made truncation so damaging in Part I.



Honest caveat. Label density differs sharply by batch (2.38 vs 1.24 labels per turn). This is annotation *granularity*, not a property of the conversations, so batch identity is carried as a covariate.

Worked examples — what a correct label set looks like

1. Empty set

“Good morning”

Strategies: none

No donation-directed content. Rapport building is reserved for *substantive* relationship work, so this is a deliberate empty set, not a missing label.

2. Single label

“i am pleasure for my charity, all the best for you”

Strategies: rapport_building

Warmth directed at the persuadee, with no information, argument, or pressure. Exactly one function is present.

3. Two labels — the removal test in action

“These organizations only ask for a small amount that we may spend on coffee each day...”

Strategies: minimization_framing, logical_appeal

Drop “small amount” and the reasoning survives; drop the coffee comparison and the minimization survives. **Both are independently licensed.**

4. Two labels — context decides

“are you sure, i mean how much money do they spend on unnecessary stuff like a bag of snack or candy?”

Strategies: comparison_framing, persistent_repetition

“are you sure” re-opens the ask *after a refusal*. **The same words earlier in the dialogue would not carry that label.**

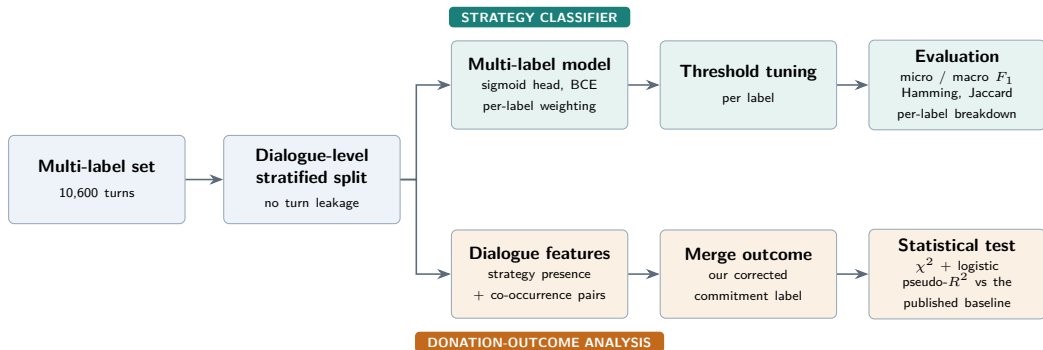
5. Four labels — a dense, layered turn

“That’s rough. I totally feel you there. I’m on short-term disability... I am not donating that assistance, I’m donating my time... I donate a few cents...”

Strategies: rapport_building, personal_story, logical_appeal, minimization_framing

Four functions, each surviving the removal test on its own. Turns with four or more labels are about 6% of the corpus — and precisely what single-label annotation destroys.

Remaining work — strategy classifier and outcome analysis



Ready to run. One notebook, single GPU, roughly half an hour. Non-neural baselines are included for comparison: prior top- k , binary relevance, and classifier chains.

The headline test. Do strategy *combinations* explain the donation outcome better than a single forced label — measured against our corrected outcome rather than the corpus field we showed to be unreliable in Part I.

Summary

Part I — delivered

1,017 dialogues re-labelled
from the text

92.21 % two-pass agreement
commitment macro-F1
0.42 → 0.84

Part II — delivered

10,600 turns annotated

41 strategies / **11** categories

48.9 % of turns are genuinely
multi-label

Next

multi-label strategy classifier
+ donation-outcome
re-analysis
on strategy *combinations*

Thank you

Questions?